Terra-Alert v3 Landslide Model: Before and After

Same model version, same 20 inputs, same app behaviour | 13 November 2026

Summary

- v3 was **memorising its training data**: it scored itself 100% on rows it had seen but 58% on new areas. After the fixes it scores 68% on seen rows and 61% on new areas, so it now learns real patterns.
- The app now **catches 70% of landslides** (the old alert level caught 41%) with **fewer false alarms** than the old model at the same catch rate (54% vs 61%).
- Safety: missing data no longer shows as "LOW risk", flat plains are always LOW, more rain can never lower the risk, and every prediction says when it should not be fully trusted.
- **Nothing the app sends or receives was removed:** the same 20 inputs, the same 0-100 score and the same LOW / MODERATE / HIGH / CRITICAL levels. v2 and v4 still work.
- **Is this final?** For the current data and these 20 inputs, yes, close to the limit. Section 6 explains why and what the next big step would be.

1. Headline numbers

All measured on **~100 km areas the model never saw during training** (nested spatial cross-validation, averaged over 3 different splits). This is the honest test: the model cannot pass it by memorising places.

	v3 before	v3 after	What it means
Ranking quality (ROC-AUC)	0.577	**0.609**	Better at putting risky places above safe ones
Landslide day vs normal day, same place	0.619	**0.665**	Better at "is today dangerous here?"
Landslide vs nearby slope, same day	0.531	**0.548**	Still hard: data too coarse (section 5)
Landslides caught at alert level	41% (score >= 40)	**70%**	Cut-off now chosen on purpose
Safe places wrongly alerted at alert level	31%	53%	More alerts: the price of catching more
Safe places wrongly alerted (at 70% caught)	61%	**54%**	Fewer false alarms
Alerts that were real (at 70% caught)	38%	**40%**	
Accuracy (at 70% caught)	50%	**54%**	
Score on its own training rows	**1.000**	**0.679**	Before: perfect memory, a red flag
Overfitting gap (training minus unseen)	**0.43**	**0.07**	Overfitting essentially gone

Why accuracy is not the main number: two of every three rows are safe, so a model that never warns would score 67% accuracy while catching zero landslides. For a warning app, **catching landslides (recall)** comes first, and false alarms are reported openly.

2. What was wrong, and what changed

#	Problem in v3 before	What changed
1	The alert level (score >= 40) was not chosen on purpose and caught only 41% of landslides. The score looked like a % chance but was not one.	Cut-offs chosen from honest test scores: MODERATE+ catches 70%, HIGH+ 40%, CRITICAL 15%. They are mapped onto the app's existing 40/60/80 bands, so no other code changed.

#	Problem in v3 before	What changed
2	Plains rule was switched off, though only 27 of 1,610 training rows are from plains. Delhi could show high risk from rain alone.	Plains rule on for v3 and v4: flat, low land (below 800 m, slope under 15 degrees) is always LOW.
3	Live rain could come from a different source than training. The backup (Open-Meteo) reports only 38-56% of the training source's rain and flipped 29% of alerts.	Training source (IMERG V07) is always tried first. Every prediction logs its source, and a backup source marks the prediction "low confidence".
3b	Missing rain or terrain data was replaced with 0, which shows as "LOW risk". An outage looked like safety.	Missing data now shows UNKNOWN (district) or an error (/predict), never a fake LOW.
5	Broken inputs: north_facing was 0 in every row (direction was averaged over 1 km, which never lands on north). Rain ratios divided by tiny numbers and reached 9,221.	These 5 inputs are still sent by the app but held constant in training, so the model ignores them.
6	Settings were copied from v2 (200 trees, depth 6). The model memorised all 1,610 rows.	40 settings tried with nested testing. Chosen: 100 short trees, each must cover at least 20 rows, plus randomness and penalties against extreme values.
-	More rain could sometimes LOWER the risk (learned noise).	Physics rule built in: more rain never lowers risk. Checked: 0 to 300 mm gives scores of 27 to 82, never dropping.
8	Data loading silently turned missing values into 0 and rewrote a CSV every time.	Missing stays missing (and is reported); no hidden file writes.
9	Test areas were ~50 km cells; one storm could sit in both training and test.	Test areas are now ~100 km blocks. Result: the score barely moved (0.578 to 0.577), so the old test was already honest.
7	Pipeline printed "ALL DONE" even after crashes.	Pipeline stops at the first failing step and says which.
-	No way to tell when a score is a guess.	Every prediction now carries *low_confidence* and *warnings*: an input outside the training range, or rain from a backup source.

3. Checked in the real backend code

Situation (same Shimla slope unless stated)	Before	After
Dry week (40 mm in 30 days)	-	28, LOW
Monsoon (60 mm in 3 days, 350 mm in 30 days)	-	65, HIGH
Cloudburst (400 mm in 3 days)	-	82, CRITICAL
Monsoon, rain from backup source (Open-Meteo)	Same score, no warning	65, HIGH + low_confidence
Delhi in monsoon (216 m, flat)	Scored by rain alone	5, LOW (plains rule)
Rain or terrain data unavailable	Treated as 0, shown as LOW	UNKNOWN / error
v2 and v4 models	Working	Still working (unchanged)

On its own 1,610 training rows, the old v3 put about 30% of rows in CRITICAL (memorised certainty). The tuned v3 puts 10% there, and most of those are real landslides.

4. Fairness check (bias), before and after

Share of landslides caught and of safe places wrongly alerted, per group, at the alert level:

Group	Caught: before	Caught: after	False alarms: before	False alarms: after
Himachal Pradesh	73%	72%	60%	**53%**
Kashmir	69%	**60%**	51%	**47%**
Uttarakhand	70%	**80%**	57%	61%
Monsoon (Jun-Sep)	73%	**79%**	58%	62%
Rest of year	63%	**44%**	48%	**29%**

The tuned model ties risk more tightly to rain, which is physically correct. The side effect: landslides with weaker rain signals are caught less often. A likely cause is other triggers such as snowmelt, but the trigger column in the catalogue has not been checked yet. This is a trade-off to watch, and it can be adjusted with region or season cut-offs (section 6). Plains states have too few rows (27) to judge, so the plains rule handles them.

5. What did not change, and what limits remain

- **Unchanged by design:** the 20 inputs, the data fetched from satellites, the 0-100 score, the four risk levels, all routes and the frontend. New response fields (*low_confidence*, *warnings*) are additions only.
- **Neighbouring slopes on the same day are still hard to separate** (ROC-AUC 0.548). Rain data comes in ~10 km squares and catalogue locations can be a few km off, so both slopes look almost identical.
- **The landslide catalogue is small, old (2007-2016) and news-based,** so remote landslides and recent disasters are missing.
- **The frontend does not show the "low confidence" badge yet:** the backend sends it, and the UI needs to display it.

6. Is this the final update?

For the current data and the same 20 inputs: yes, this is close to the limit. In the tuning step, 40 very different model settings all landed around ROC-AUC 0.60-0.61 on unseen areas. When very different models agree, the limit is the information in the data, not the model. More tuning would not move it meaningfully, and pushing harder would only bring overfitting back.

What could still improve things, from smallest to largest:

Option	Keeps the 20 inputs?	Expected effect
Show the "low confidence" badge in the app	Yes	Users know when to trust a score less
Separate cut-offs per region or season	Yes	Can even out the Kashmir / non-monsoon catch rate; does not make the model smarter
Better landslide data (Geological Survey of India, Bhukosh inventory)	Yes	**Largest gain.** More, more precise and more recent events. Must be downloaded by a team member
Small-scale terrain inputs (tested earlier)	No (adds inputs)	Tested: +0.02 to +0.03 ROC-AUC. Real but small
Two-part model: "where" map x "when" rain trigger	No (new design)	How professional systems work; the best route to clearly higher, honest scores

Honest expectation: 80%+ accuracy with good recall is not reachable with this data without cheating. The older v2 showed 90% only because of data leakage (its safe examples were plains and glaciers) and scored worse than a coin flip on a fair test.

Glossary

Term	Meaning
Recall (caught)	Of all real landslides, the share the model warned about.
False-alarm rate	Of all safe places/days, the share that wrongly got a warning.
Precision	Of all warnings, the share that were real landslides.
ROC-AUC	How well the model ranks risky above safe. 0.5 = coin flip, 1.0 = perfect.
Overfitting	Memorising training rows instead of learning patterns: great on seen data, weak on new data.
Nested cross-validation	Settings are chosen using only training areas; a separate set of untouched areas gives the reported score, so the score is not flattered by the choice.
Monotone constraint	A rule built into the model: more rain can only keep or raise the risk.

Scripts: ai-module/v3/tune_v3.py, train_xgboost_v3.py, calibrate_thresholds_v3.py, compare_rain_sources_v3.py. Previous model backed up in ai-module/v3/backup_before_tuning_2026-11-13/.